



Heart Sound Classification

Speech Processing Project

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Introduction

Heart sounds are recordings of heartbeats.

They contain important information about how the heart works.

Doctors use these sounds to detect heart problems.

Today, machine learning can analyze heart sounds automatically and help detect abnormalities faster and more accurately.

In this project, we used:

- Audio signal processing
- Machine learning models

to classify heart sounds into:

- Normal
- Abnormal

Problem Definition

Classify heart sound recordings as Normal or Abnormal

Input

Heart sound audio files (.wav)

Output

Prediction result (Normal / Abnormal)

Main Challenges

Noise in recordings, different recording environments, and small differences between sounds.

Why Is This Important?

Early Detection

Heart diseases can be detected early from abnormal sounds.

Better Access

Helps doctors in places without specialists.

Faster Diagnosis

Automatic systems save time.

More Accuracy

Reduces human errors.

Applications

Hospitals

Helping doctors diagnose patients

Mobile Apps

Monitoring patients remotely

Smart Devices

Digital stethoscopes

Telemedicine

Remote medical diagnosis

Education

Training medical students

Dataset

Dataset used:

PhysioNet Heart Sound Dataset (2016)

Source : [Classification of Heart Sound Recordings: The PhysioNet/Computing in Cardiology Challenge 2016 v1.0.0](#)

Total recordings:

About 3240

Classes:

- Normal
- Abnormal

This is a:

Binary classification problem

Dataset Characteristics

Sampling Rate

16000 Hz

Duration

35.67 seconds

Class

Normal , Abnormal

Class Distribution

Abnormal 2831 Normal 409

after augmentation Abnormal 2831 Normal 1636

Number of Samples

570656

Preprocessing

Before training the model, the audio was cleaned and prepared.

Noise Removal

A band-pass filter (20-200 Hz) removes breathing noise and electrical interference, preserving only heart sound frequencies.

Silence Removal

Silent/low-energy segments are removed using RMS thresholding, keeping only informative heart sound regions.

Augmentation

3 augmented copies are generated per normal file (Time Stretch, Pitch Shift, Noise, Shift) to balance the dataset from 1,000 to 4,000 normal samples.

Normalization

The signal is normalized to the range $[-1, 1]$ to ensure consistent loudness across all recordings.

Segmentation

Recordings are truncated to a maximum of 10 seconds, enough for 10-20 heart cycles.

Feature Extraction

The computer cannot understand raw audio directly. So we convert sound into numbers called features.



Represents sound patterns.

**Delta MFCC**

Captures how MFCC changes over time

**Spectral Contrast**

Compares peaks and valleys in the frequency spectrum

**Spectral Features**

Describe frequency information.

**Delta2 MFCC**

Captures acceleration of MFCC changes

**Tonnetz**

Describes harmonic relationships in the sound

**Statistical Features**

Describe signal behavior.

**Chroma Features**

Describes musical note distribution

**Zero Crossing Rate****(ZCR)**

Measures how often the signal changes from positive to negative

**RMS Energy**

Measures the average signal power (actual loudness)

**Spectral Centroid**

The "center of gravity" of the frequency spectrum.

Feature Visualization

We used graphs to understand the heart sound signals and check preprocessing

Waveform

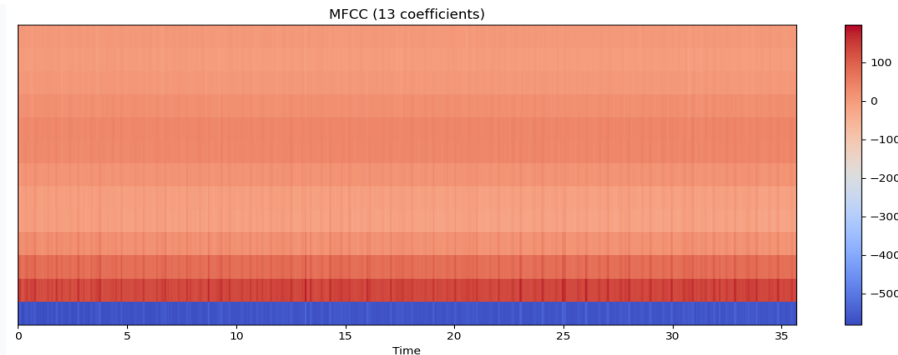
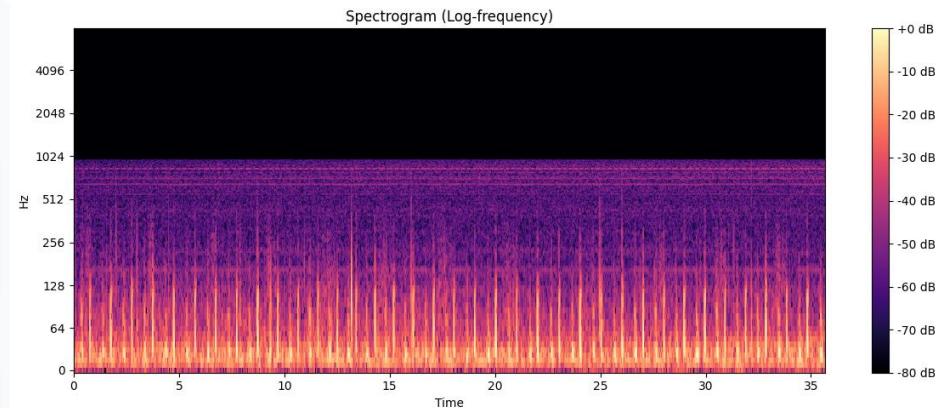
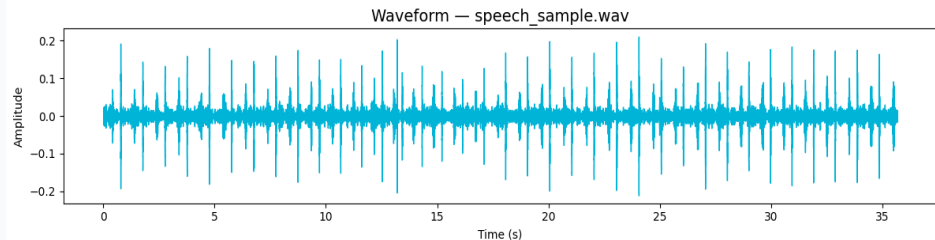
Shows sound amplitude over time.

Spectrogram

Shows frequency and energy.

MFCC Plot

Shows important sound features.



Model

We used machine learning models to classify heart sounds.

Models used:

- XGBoost
- LightGBM
- CNN

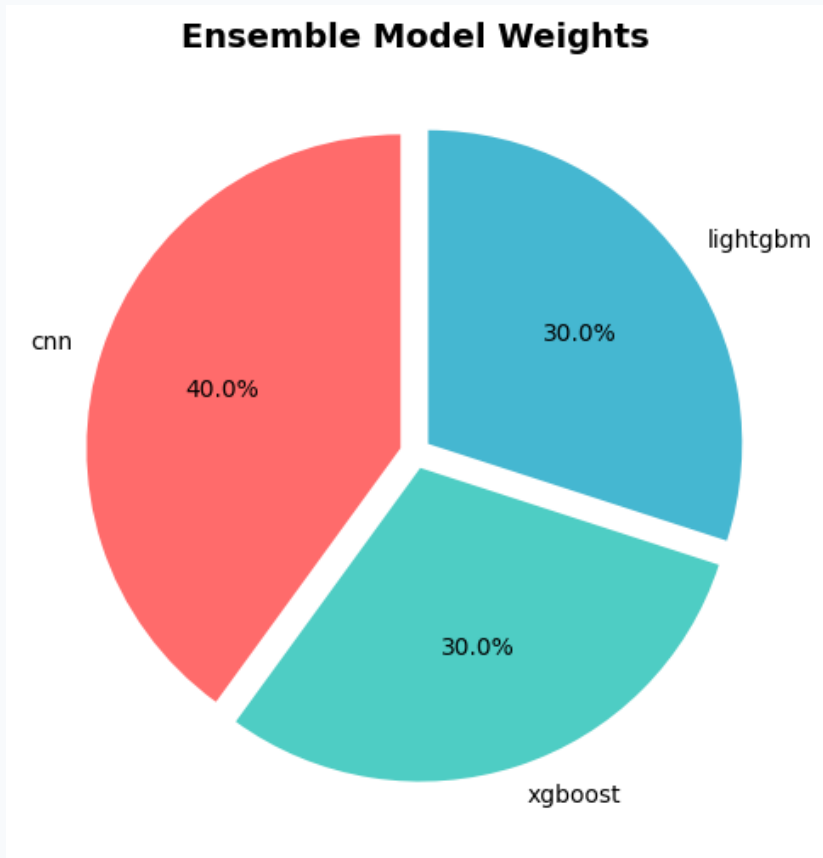
These models learn patterns from the extracted features.

Process:

Audio → Preprocessing → Feature Extraction
→ Model → Prediction

Output:

Normal or Abnormal



Training

Data Split

80% Training, 20% Testing

Hardware

Standard computer (CPU)

Loss Function

Binary Log Loss

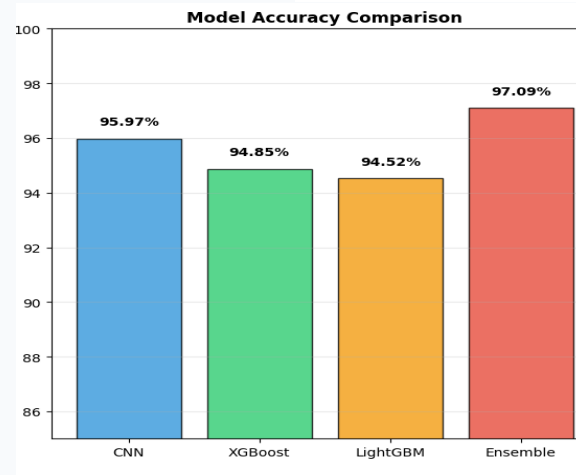
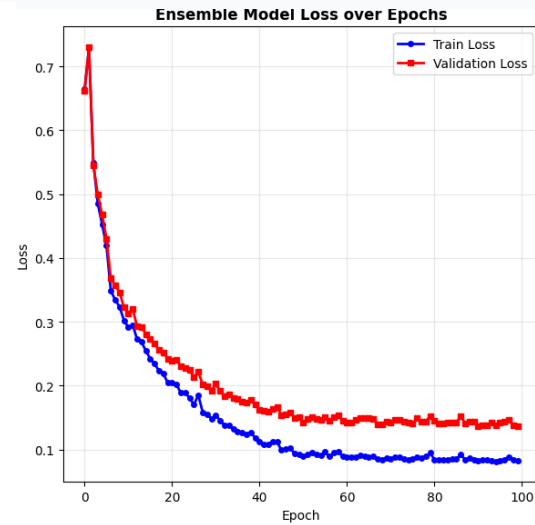
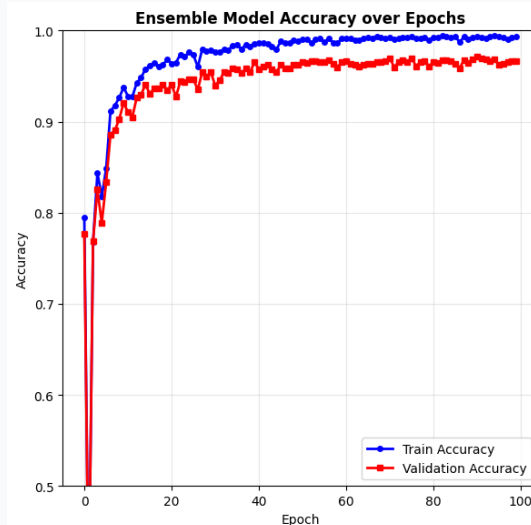
Note

⚙️ CNN Specific Parameters

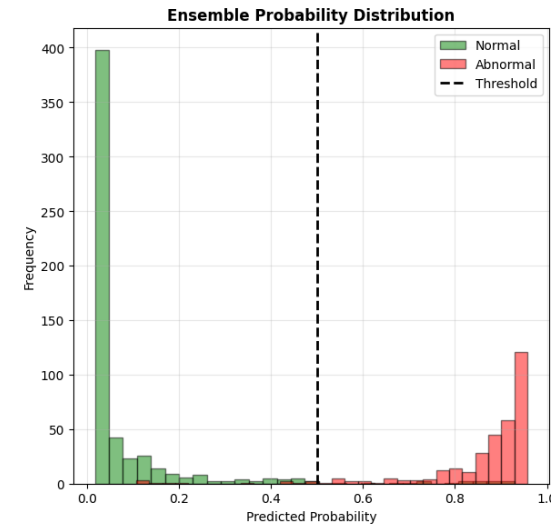
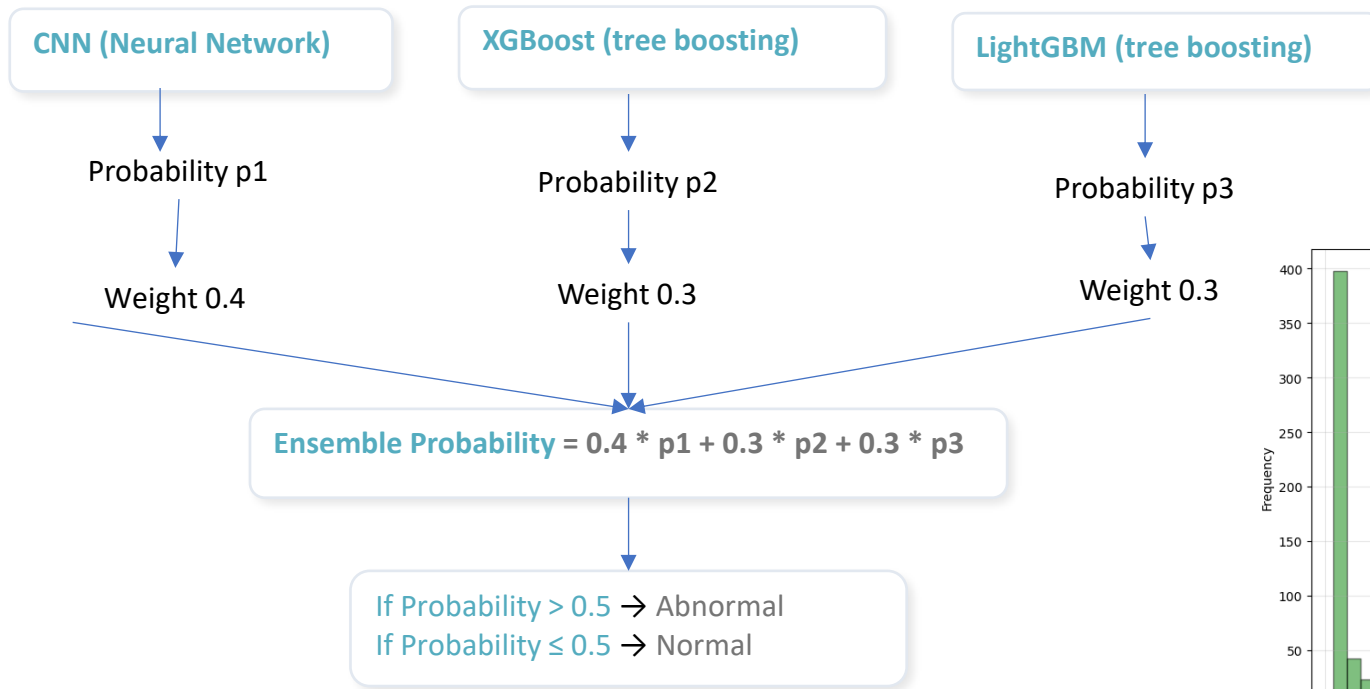
- Epochs: 100 (number of full training passes)
- Batch Size: 32 (samples per training step)

🌲 XGBoost & LightGBM Specific Parameters

- n_estimators: 200 trees
- max_depth: 6 levels per tree
- learning_rate: 0.05



Ensemble Model



Results

We evaluated the model using various metrics to ensure robustness.

Accuracy

Overall correctness of the model. 97.09

Recall

How many real abnormal cases were successfully detected. 0.9664

Confusion Matrix

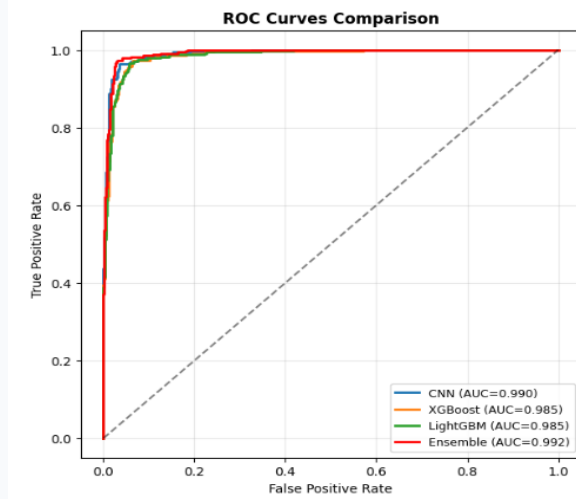
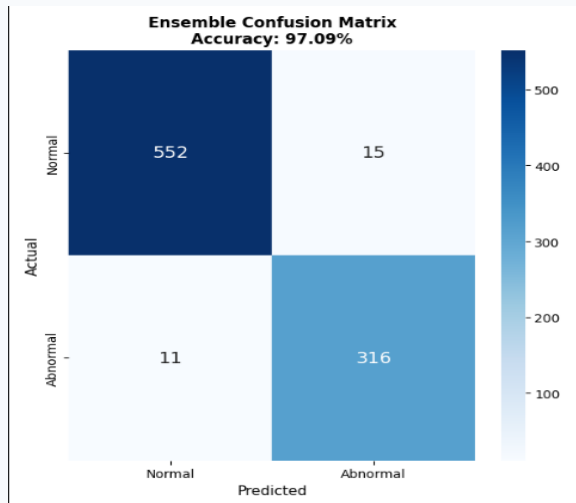
Shows detailed prediction errors.

Precision

How many predicted abnormal cases are actually correct. % 0.9547

F1-score

Balance between precision and recall. 0.9605



Discussion

What Worked Well

Good feature extraction, clean preprocessing, and using more than one model improved accuracy.

Challenges

Noisy recordings, imbalanced dataset, and different recording lengths.

Model Performance

The model performed well and generalized successfully to new data.

Conclusion

This project shows that machine learning can successfully classify heart sounds.

The System Can

Detect abnormal heart sounds, improve diagnosis speed, and support medical applications.

Future Work

Use larger datasets, build real-time systems, and integrate with medical devices.



Thank You